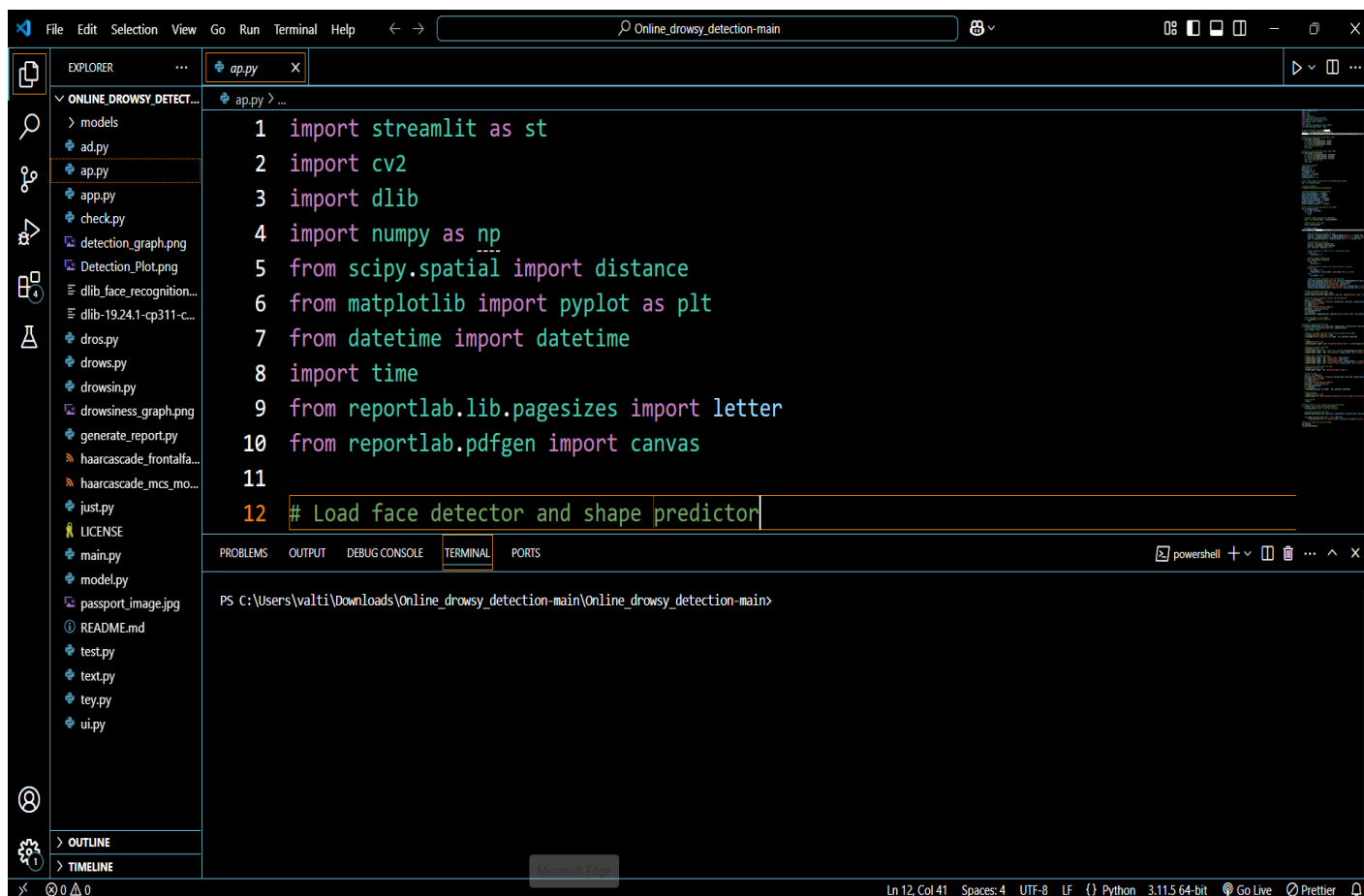


APPENDICES

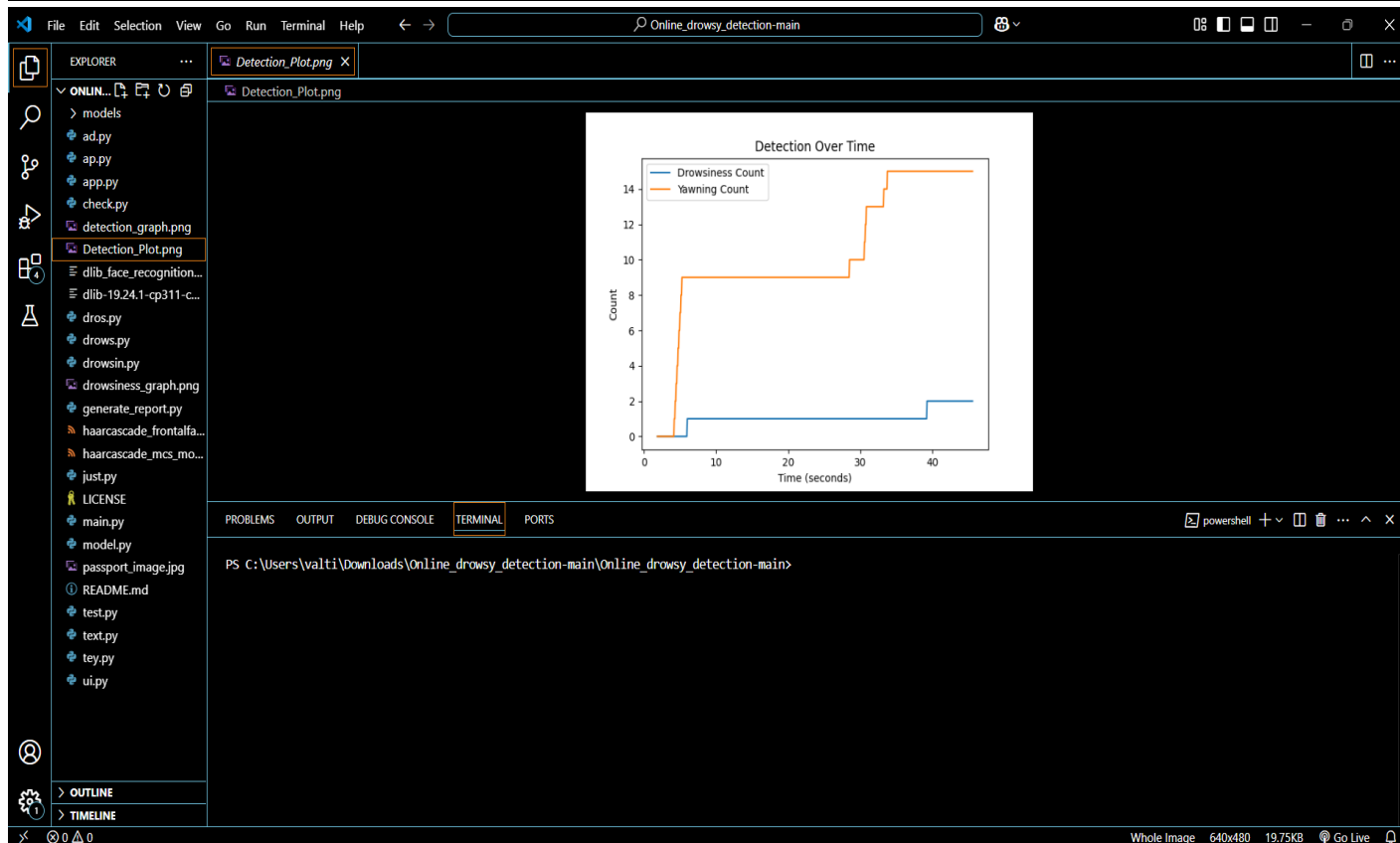
PROGRAMING AND RUNNING ATTACHMENTS:



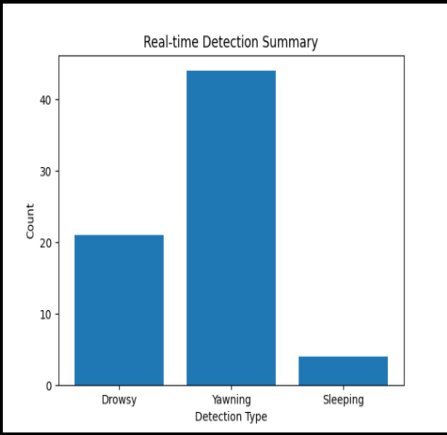
The screenshot shows the Visual Studio Code editor with the file 'ap.py' open. The code is a Python script for online drowsy detection. The Explorer sidebar on the left shows the project structure, including files like 'ad.py', 'app.py', 'check.py', 'detection_graph.png', 'Detection_Plot.png', 'dlib_face_recognition...', 'dlib-19.24.1-cp311-c...', 'drows.py', 'drowsin.py', 'drowsiness_graph.png', 'generate_report.py', 'haarcascade_frontalface...', 'haarcascade_mcs_...', 'just.py', 'LICENSE', 'main.py', 'model.py', 'passport_image.jpg', 'README.md', 'test.py', 'text.py', 'tey.py', and 'ui.py'. The code in 'ap.py' is as follows:

```
1 import streamlit as st
2 import cv2
3 import dlib
4 import numpy as np
5 from scipy.spatial import distance
6 from matplotlib import pyplot as plt
7 from datetime import datetime
8 import time
9 from reportlab.lib.pagesizes import letter
10 from reportlab.pdfgen import canvas
11
12 # Load face detector and shape predictor
```

The terminal at the bottom shows the command prompt 'PS C:\Users\valti\Downloads\Online_drowsy_detection-main\Online_drowsy_detection-main>'.



VS Code interface showing the Explorer view on the left with a file tree. The main editor displays a bar chart titled "Real-time Detection Summary". The chart has a Y-axis labeled "Count" ranging from 0 to 40 and an X-axis labeled "Detection Type" with categories "Drowsy", "Yawning", and "Sleeping". The bars show counts of approximately 21 for Drowsy, 45 for Yawning, and 4 for Sleeping.



Detection Type	Count
Drowsy	21
Yawning	45
Sleeping	4

The bottom panel shows the TERMINAL view with the command prompt: PS C:\Users\valti\Downloads\Online_drowsy_detection-main\Online_drowsy_detection-main>

VS Code interface showing the Explorer view on the left with a file tree. The main editor displays the code for main.py. The code includes imports for streamlit, cv2, dlib, numpy, scipy, matplotlib, datetime, time, reportlab, and canvas. It also includes a comment indicating the loading of a face detector and shape predictor.

```
1 import streamlit as st
2 import cv2
3 import dlib
4 import numpy as np
5 from scipy.spatial import distance
6 from matplotlib import pyplot as plt
7 from datetime import datetime
8 import time
9 from reportlab.lib.pagesizes import letter
10 from reportlab.pdfgen import canvas
11
12 # Load face detector and shape predictor
```

The bottom panel shows the TERMINAL view with the command prompt: PS C:\Users\valti\Downloads\Online_drowsy_detection-main\Online_drowsy_detection-main>

IMPLEMENTATION ATTACHMENTS:

ap

localhost:8501

RUNNING... Stop Deploy

Drowsiness Detection Dashboard

Start Time: 2025-03-10 01:42:15

Drowsy Count: 3

Yawning Count: 0

Total Sleep Duration: 0.19 minutes

Screen On Time: 0.00 minutes



ap

localhost:8501

RUNNING... Stop Deploy

Passport Image

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the use_container_width parameter instead.

12

Load face detector and shape predictor

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell + - [] ... ^ x

2025-03-10 01:45:05.856 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:05.907 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:06.054 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:06.113 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:06.202 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:06.270 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:06.344 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:06.412 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:06.489 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:06.556 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:06.634 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:06.708 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
2025-03-10 01:45:06.776 The 'use_column_width' parameter has been deprecated and will be removed in a future release. Please utilize the 'use_container_width' parameter instead.
Stopping...

PS C:\Users\valti\Downloads\Online drowsy detection-main\Online drowsy detection-main>

ap localhost:8501

... CONNECTING

Drowsiness Detection Dashboard

Start Time: 2025-03-10 01:42:15

Drowsy Count: 22

Yawning Count: 33

Total Sleep Duration: 2.82 minutes

Screen On Time: 0.00 minutes